1 EXAMPLE

EXA MPLE

E XA MP LE

E X A M P L E

E AX MP EL

AEX ELMP

AEELMPX

2.i) First element as pivot.

ABCDE 4 with i
↑ i j 4 with j
ABCDE
↑ j i

BCDE 3 with i
↑ i j 3 with j
CDE 2 with i
↑ i j 2 with j

DE 1 with i ✗
↑ i j 1 with j ✗
E 0 comp

~~[illegible] comparisons.~~

8 + 6 + 4 + 0 = 18 comparisions.

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ii) Last element as pivot,

A B C D E
i i ↑ j

increment i till i^{th} pos is greater than pivot
Decrement j till j^{th} pos
j^{th} is less than pivot

A B C D E 5+1 ← with j
  j i ↑ with i

A B C D
i ↑ j

A B C D 4+1
  j ↑ i

A B C 3+1
i i ↑ j

18 Comparisions.

A B 2+1
i j

Swap.

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3.

$$\begin{array}{c} \begin{array}{cc} a_{00} & a_{01} \end{array} \\ \begin{bmatrix} \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} & \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} \\ \begin{bmatrix} 0 & 1 \\ 5 & 0 \end{bmatrix} & \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix} \end{bmatrix} \\ \begin{array}{cc} a_{10} & a_{11} \end{array} \end{array} \times \begin{array}{c} \begin{array}{cc} b_{00} & b_{01} \end{array} \\ \begin{bmatrix} \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix} & \begin{bmatrix} 0 & 1 \\ 0 & 4 \end{bmatrix} \\ \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} & \begin{bmatrix} 1 & 1 \\ 5 & 0 \end{bmatrix} \end{bmatrix} \\ \begin{array}{cc} b_{10} & b_{11} \end{array} \end{array}$$

$m_1 = (a_{00} + a_{11}) * (b_{00} + b_{11})$

$$\begin{bmatrix} 4 & 0 \\ 6 & 2 \end{bmatrix} \times \begin{bmatrix} 0 & 2 \\ 2 & 8 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 7 & 1 \end{bmatrix}$$

$m_1 = 12$ $m_2 = 8$ $m_3 = 4$, $m_4 = 12$ $m_5 = 4$

$m_6 = 6$ $m_7 = -16$

$$m_1 = \begin{bmatrix} 4 & 8 \\ 20 & 14 \end{bmatrix}.$$

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$m_2 = (a_{10} + a_{11}) * b_{00}$

$$\begin{bmatrix} 3 & 1 \\ 7 & 1 \end{bmatrix} * \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}$$

$m_1 = 4$ $m_2 = 0$ $m_3 = 0$ $m_4 = 2$ $m_5 = 4$ $m_6 = 4$
$m_7 = 0$

$$m_2 = \begin{bmatrix} 2 & 4 \\ 2 & 8 \end{bmatrix}$$

$m_3 = a_{00} * (b_{01} - b_{11})$

$$\begin{bmatrix} 0 & 1 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 1 & 1 \\ 5 & 0 \end{bmatrix}$$

$$\begin{matrix} 1 & 0 \\ 4 & 1 \end{matrix} * \begin{bmatrix} -1 & 0 \\ -5 & 4 \end{bmatrix}$$

$m_1 = 6$ $m_2 = -5$ $m_3 = -4$, $m_4 = -4$ $m_5 = 4$
$m_6 = -3$ $m_7 = 1$

$$m_3 \begin{bmatrix} -1 & 0 \\ -9 & 4 \end{bmatrix}$$

$m_4 = a_{11} * (b_{10} - b_{00})$

$$= \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

$m_1 = 16$ $m_2 = 6$ $m_3 = -9$ $m_4 = -3$ $m_5 = 6$ $m_6 = -1$
$m_7 = -1$

$$m_4 = \begin{bmatrix} 6 & -3 \\ 3 & 0 \end{bmatrix}$$

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$m_5 = (a_{00} + a_{01}) * b_{11}$

$$\begin{bmatrix} 3 & 1 \\ 5 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 1 \\ 5 & 0 \end{bmatrix}$$

$m_1 = 4$ $m_2 = 6$ $m_3 = 3$ $m_4 = 4$ $m_5 = 0$ $m_6 = 4$
$m_7 = 0$

$$m_5 = \begin{bmatrix} 8 & 3 \\ 10 & 5 \end{bmatrix}$$

$m_7 = (a_{10} - a_{00}) * (b_{00} + b_{01})$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} * \begin{bmatrix} 0 & 2 \\ 2 & 5 \end{bmatrix}$$

$m_1 = -10$ $m_2 = 0$ $m_3 = 3$ $m_4 = -2$ $m_5 = 0$ $m_6 = 4$
$m_7 = 14$

$$m_6 = \begin{bmatrix} 2 & 3 \\ -2 & 3 \end{bmatrix}$$

$m_7 = (a_{01} - a_{11}) * (b_{10} + b_{11})$

$$\begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix} * \begin{bmatrix} 3 & 1 \\ 6 & 3 \end{bmatrix}$$

$m_1 = -12$ $m_2 = -6$ $m_3 = 2$ $m_4 = -3$ $m_5 = 0$ $m_6 = 0$
$m_7 = 18$

$$m_7 = \begin{bmatrix} 3 & 2 \\ -9 & -4 \end{bmatrix}$$

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$$m_3 + m_5 = \begin{bmatrix} -1 & 0 \\ -9 & 4 \end{bmatrix} + \begin{bmatrix} 8 & 3 \\ 10 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 7 & 3 \\ 1 & 9 \end{bmatrix}$$

$$m_2 + m_4 = \begin{bmatrix} 2 & 4 \\ 2 & 8 \end{bmatrix} + \begin{bmatrix} 6 & -3 \\ 3 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 1 \\ 5 & 8 \end{bmatrix}$$

$m_1 + m_4 - m_5 + m_7 =$

$$\begin{bmatrix} 4 & 8 \\ 20 & 14 \end{bmatrix} + \begin{bmatrix} 6 & -3 \\ 3 & 0 \end{bmatrix} - \begin{bmatrix} 8 & 3 \\ 10 & 5 \end{bmatrix} + \begin{bmatrix} 3 & 2 \\ -9 & -4 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$$

$m_1 + m_3 - m_2 + m_6$

$$\begin{bmatrix} 4 & 8 \\ 20 & 14 \end{bmatrix} + \begin{bmatrix} -1 & 0 \\ -9 & 4 \end{bmatrix} - \begin{bmatrix} 2 & 4 \\ 2 & 8 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ -2 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 7 \\ 7 & 7 \end{bmatrix}$$

$$C = \begin{bmatrix} 5 & 4 & 7 & 3 \\ 4 & 5 & 1 & 9 \\ 8 & 1 & 3 & 7 \\ 5 & 8 & 7 & 7 \end{bmatrix}$$

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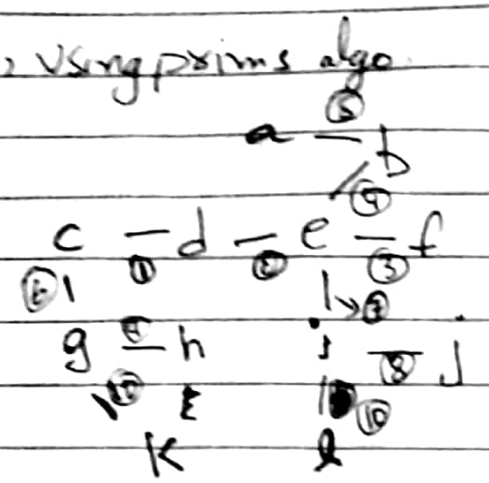
4.

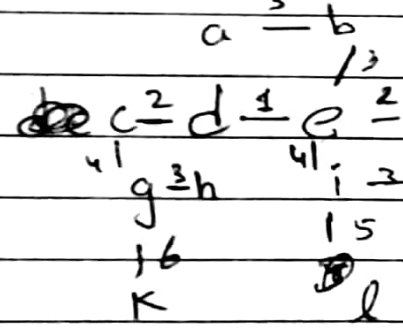
i) Using prims algo.



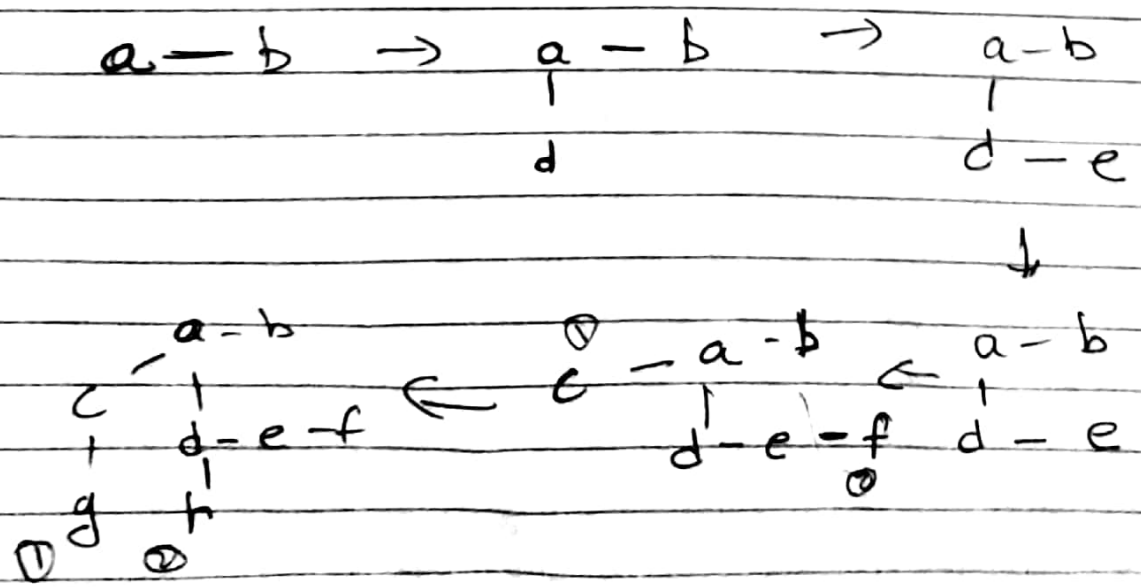
ii) Using Kruskal's algorithm.

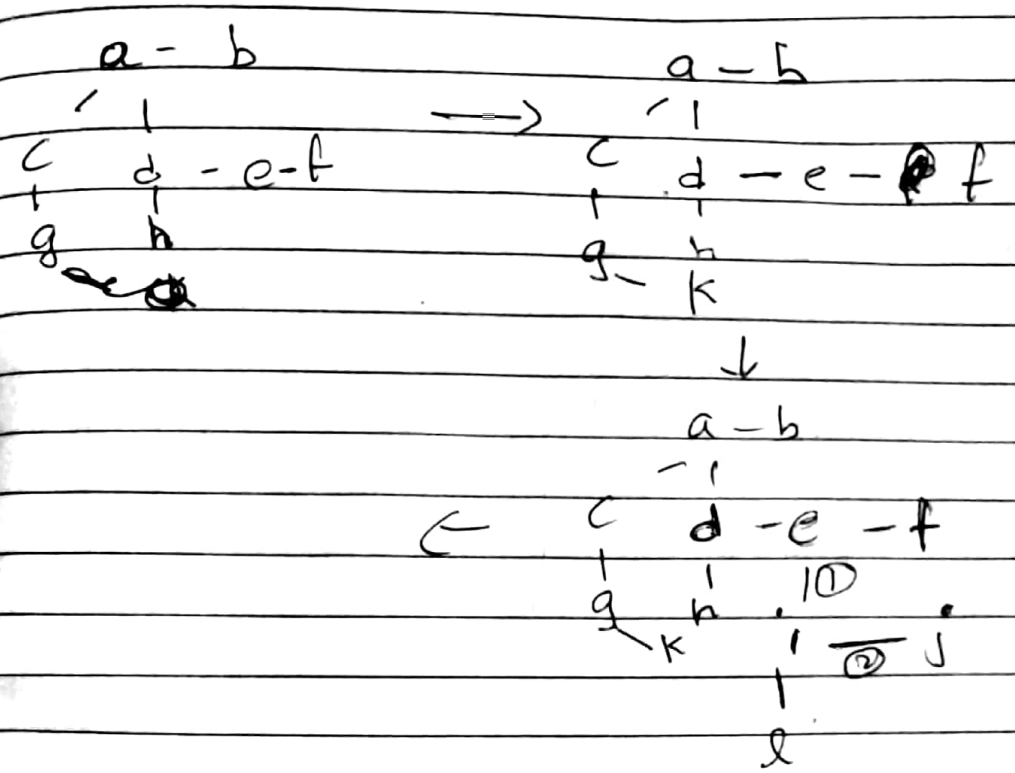
✓	✓	✓	✓	✓	✓	✓	✗	✓	✓
1	2	2	3	3	3	3	4	4	4
de	cd	ef	ab	be	gh	ij	ad	cg	ei

5	5	5	5	6	6	6	7	8	9
ca	dh	fj	il	bf	hi	gk	hk	kl	lj
✗	✗	✗		✗	✗	✓	✗	✗	✗



5. Source vertex is as 'a' Dijkstra's





1.

	i \ j	0	1	2	3	4	5	6
	0	0	0	0	0	0	0	0
w=3 v=25	1	0	0	0	25	25	25	25
w=2 v=20	2	0	0	20	25	25	45	45
w=1 v=15	3	0	15	20	35	40	45	60
w=4 v=40	4	0	15	20	35	40	55	60
w=5 v=50	5	0	15	20	35	40	55	65

45
15
60

Item 4 included. set the weight capacity → 5
Item 3 included.

2. $R^0 = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$

$R_1 = \begin{array}{c} \\ \\ \\ a \end{array} \begin{array}{c} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$

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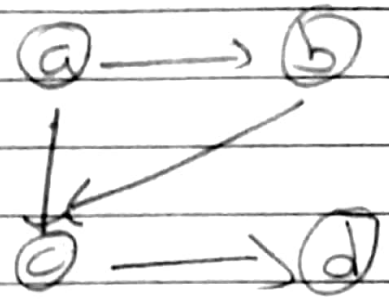
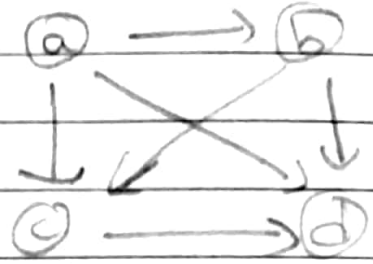
$$R_2 = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

0, a, b



$$R_3 = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

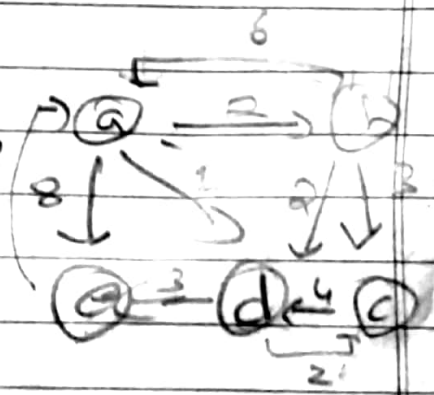
0, a, b, c



$$R_4 = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

0, a, b, c, d

3. $$R^0 = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \end{array} \begin{array}{c} \begin{array}{ccccc} a & b & c & d & e \end{array} \\ \begin{bmatrix} 0 & 2 & \infty & 1 & 8 \\ 6 & 0 & 3 & 2 & \infty \\ \infty & \infty & 0 & 4 & \infty \\ \infty & \infty & 2 & 0 & 3 \\ 3 & \infty & \infty & \infty & 0 \end{bmatrix} \end{array}$$



$$R_1 = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \end{array} \begin{array}{c} \begin{array}{ccccc} a & b & c & d & e \end{array} \\ \begin{bmatrix} 0 & 2 & \infty & 1 & 8 \\ 6 & 0 & 3 & 2 & 14 \\ \infty & \infty & 0 & 4 & \infty \\ \infty & \infty & 2 & 0 & 3 \\ 3 & 5 & \infty & 4 & 0 \end{bmatrix} \end{array}$$

0, a

R_0 : 0, a, b.

$$\begin{array}{c|ccccc} & a & b & c & d & e \\ \hline a & 0 & 2 & 5 & 1 & 8 \\ b & 6 & 0 & 3 & 2 & 14 \\ c & \infty & \infty & 0 & 4 & \infty \\ d & \infty & \infty & 2 & 0 & 3 \\ e & 3 & 5 & 8 & 4 & 0 \end{array}$$

R_3 : 0, a, b, c

$$\begin{array}{c|ccccc} & a & b & c & d & e \\ \hline a & 0 & 2 & 5 & 1 & 8 \\ b & 6 & 0 & 3 & 2 & 14 \\ c & \infty & \infty & 0 & 4 & \infty \\ d & \infty & \infty & 2 & 0 & 3 \\ e & 3 & 5 & 8 & 4 & 0 \end{array}$$

R_4 : 0, a, b, c, d

$$\begin{array}{c|ccccc} & a & b & c & d & e \\ \hline a & 0 & 2 & 3 & 1 & 4 \\ b & 6 & 0 & 3 & 2 & 5 \\ c & \infty & \infty & 0 & 4 & 7 \\ d & \infty & \infty & 2 & 0 & 3 \\ e & 3 & 5 & 6 & 4 & 0 \end{array}$$

R_5 : 0, a, b, c, d, e

$$\begin{array}{c|ccccc} & a & b & c & d & e \\ \hline a & 0 & 2 & 3 & 1 & 4 \\ b & 6 & 0 & 3 & 2 & 5 \\ c & 10 & 12 & 0 & 4 & 7 \\ d & 6 & 8 & 2 & 0 & 3 \\ e & 3 & 5 & 6 & 4 & 0 \end{array}$$

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